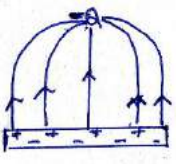
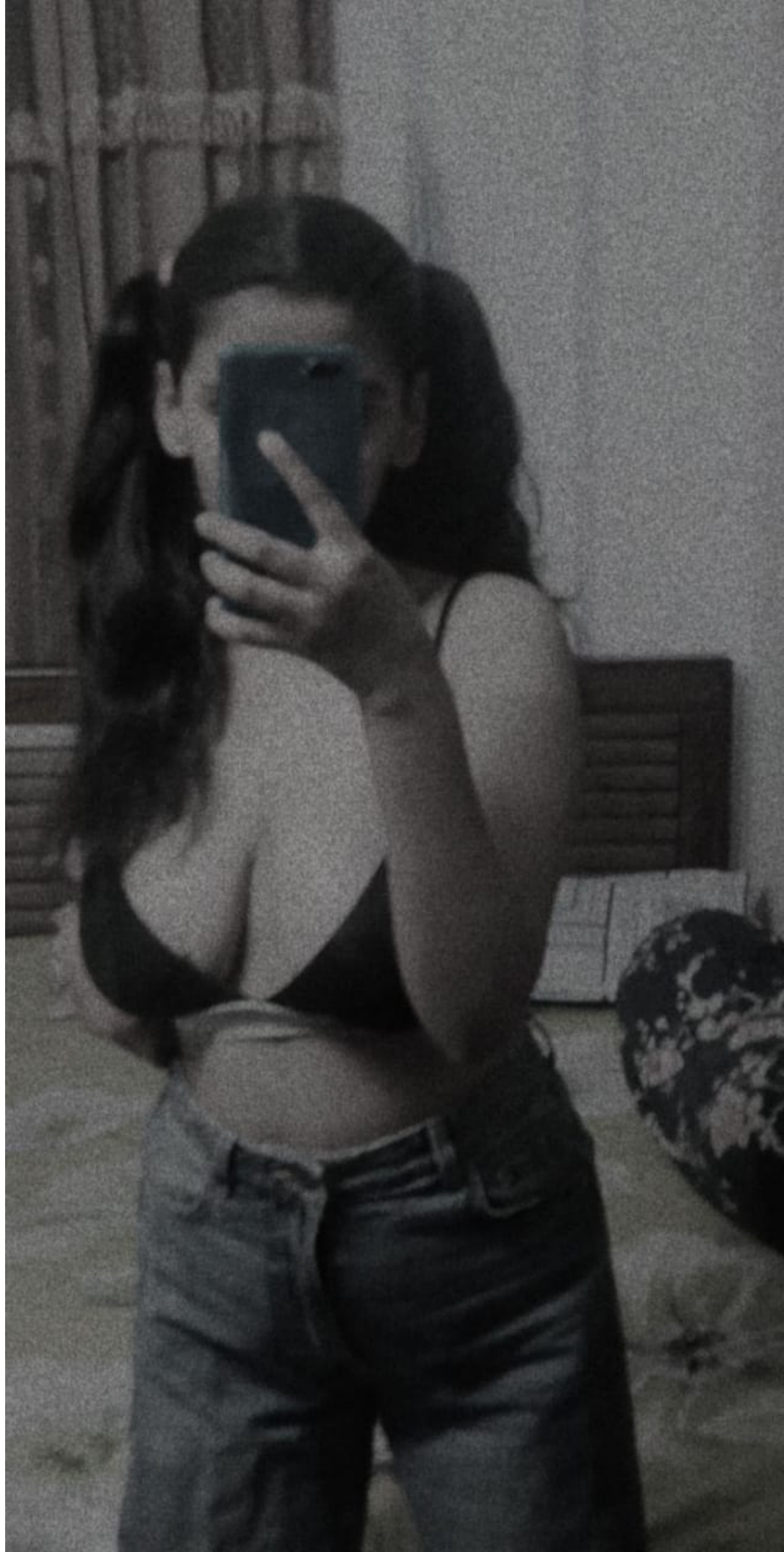


CBSE Class 12 Physics Question Paper Solution

55/1/1

MARKING SCHEME – PHYSICS

Q. No.	Value Points/ Expected answers	Marks	Total Marks
1	 [Note: i) Deduct ½ mark, if arrows are not shown. ii) do not deduct any mark, if charges on the plates are not shown]	1	1
2	No Change	1	1
3	Threshold frequency equals the minimum frequency of incident radiation (light) that can cause photoemission from a given photosensitive surface. (Alternatively) The frequency below which the incident radiations cannot cause the photoemission from photosensitive surface. OR Intensity of radiation is proportional to (/ equal to) the number of energy quanta (photons) per unit area per unit time.	1	1
4	$d\mu_r = \tan 30^\circ = \frac{1}{\sqrt{3}}$ (where $d\mu_r$ is the refractive index of rarer medium w.r.t denser medium) $\therefore \mu_d = \sqrt{3}$ $v = \frac{c}{\mu} = \frac{3 \times 10^8}{\sqrt{3}} = \sqrt{3} \times 10^8 \text{ m/s}$ [Note- Also accept if a student solves it as follows] $\mu = \tan i_p$ $\mu = \tan 30^\circ = \frac{1}{\sqrt{3}}$ $\therefore v = \frac{3 \times 10^8}{\frac{1}{\sqrt{3}}} = 3\sqrt{3} \times 10^8 \text{ m/s}$ (Note: Award this one mark if a student just writes the formula but does not solve it.)	½ ½ ½ ½	1
5	The waves beyond 30 MHz frequency penetrate through the Ionosphere/ are not reflected back. OR Transmitted Power and Frequency	1 ½ + ½	1
SECTION - B			
6	Calculation of Power dissipation in two combinations 1 +1 $R_1 = \frac{V^2}{P_1}, R_2 = \frac{V^2}{P_2},$ $P_s = \frac{V^2}{R_s} = \frac{P_1 P_2}{P_1 + P_2}$ $\frac{1}{P_s} = \frac{1}{P_1} + \frac{1}{P_2}$ $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{P_1 + P_2}{V^2}$	½ ½ ½	



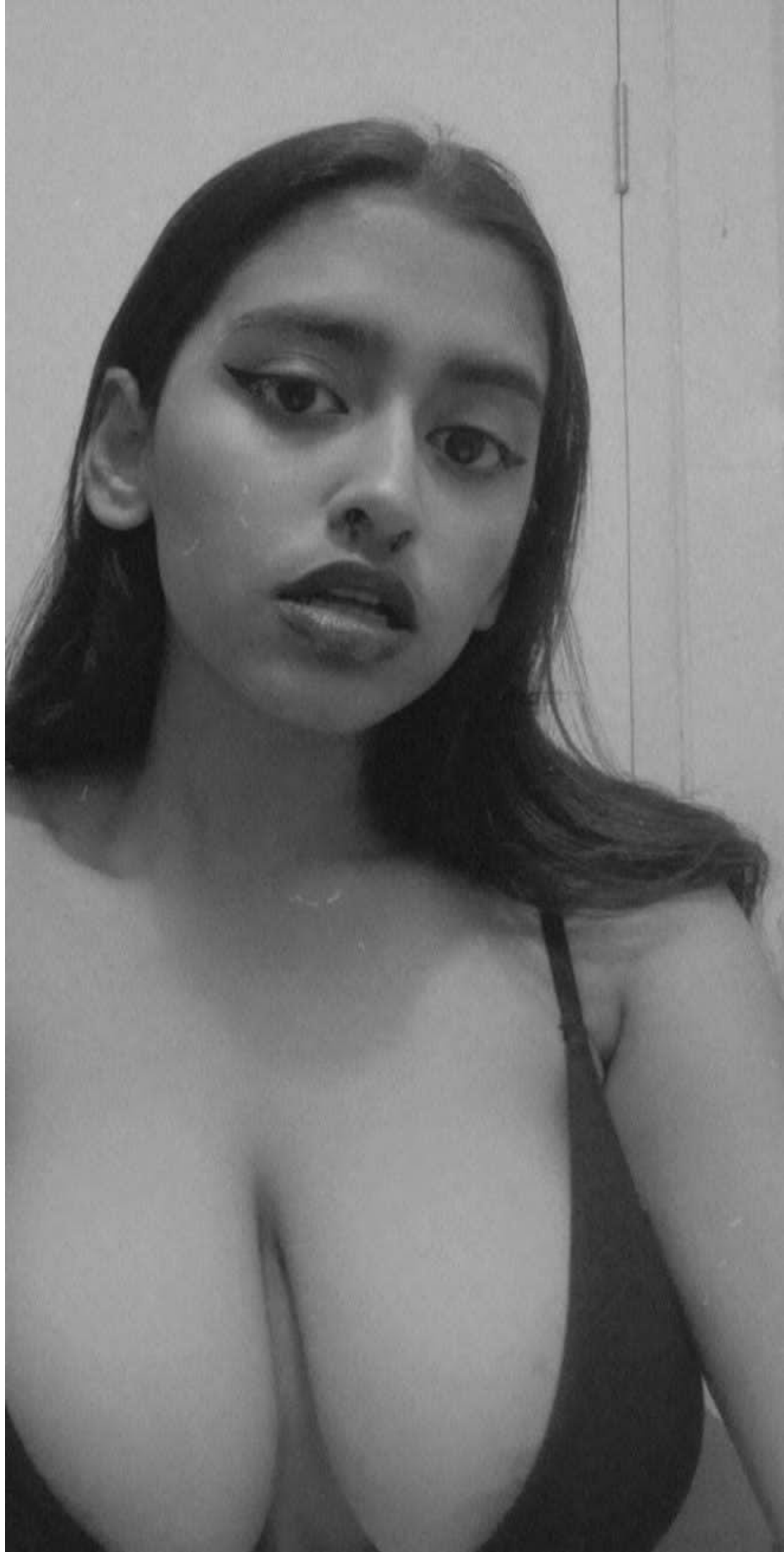








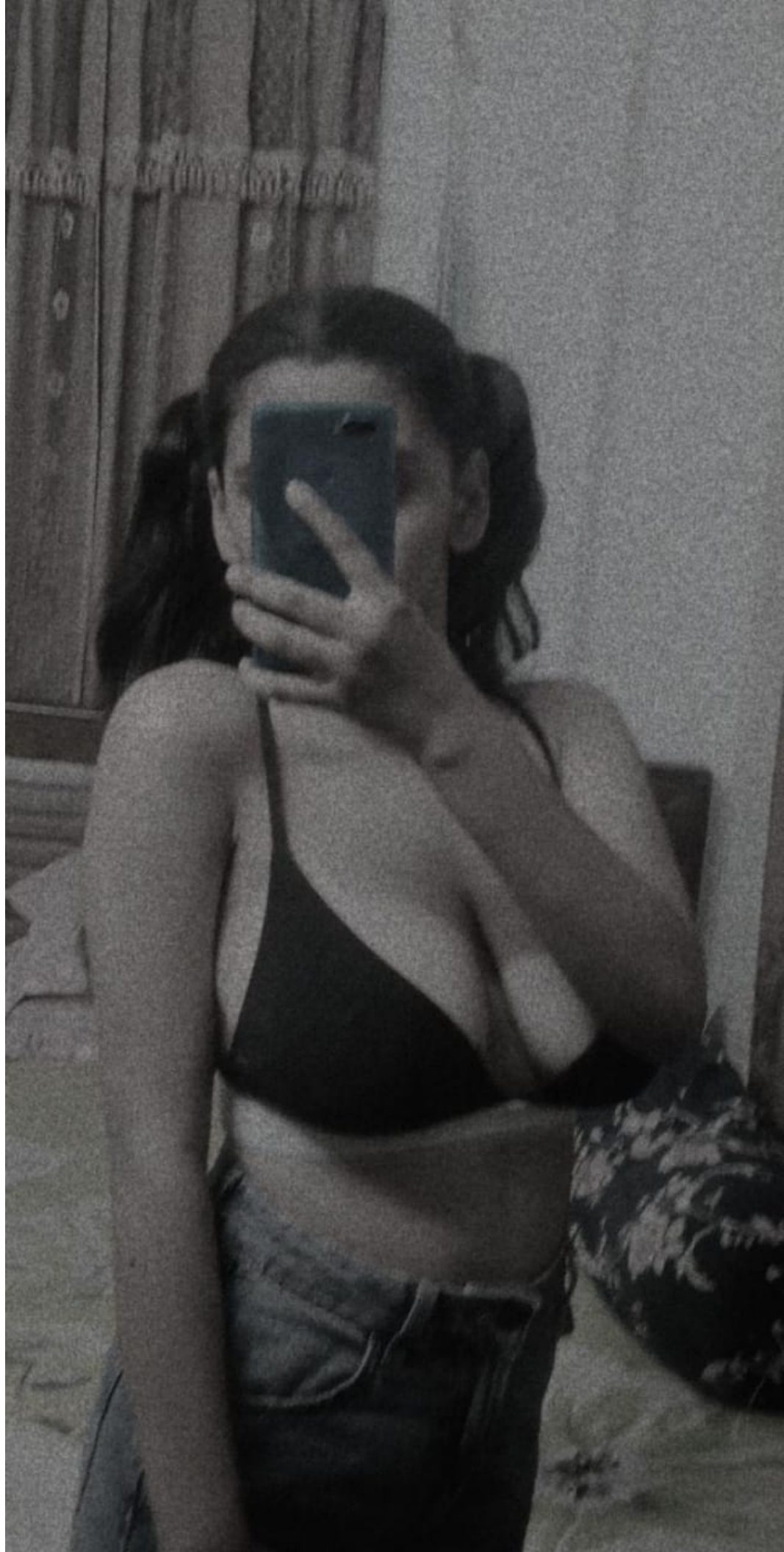




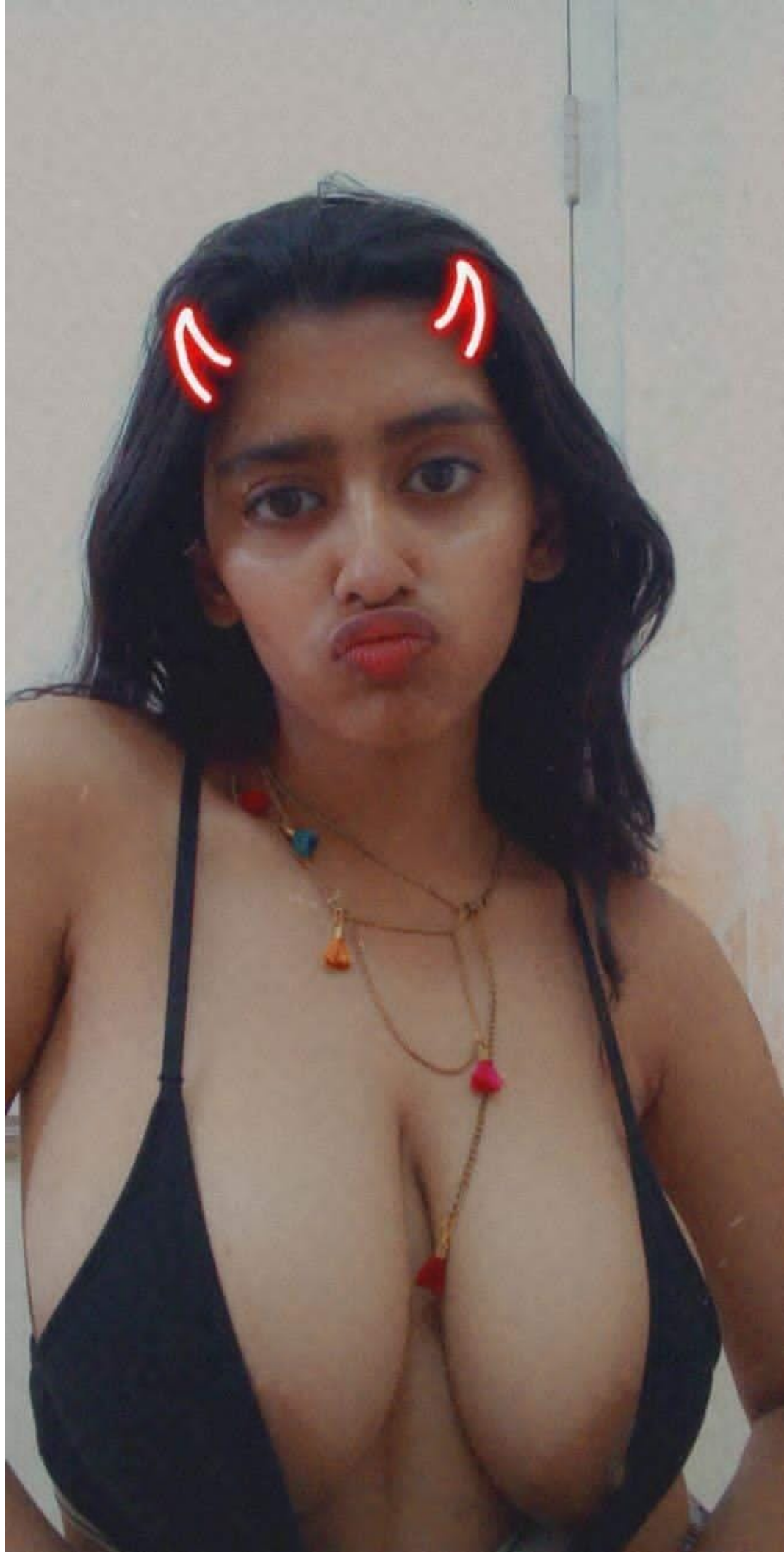


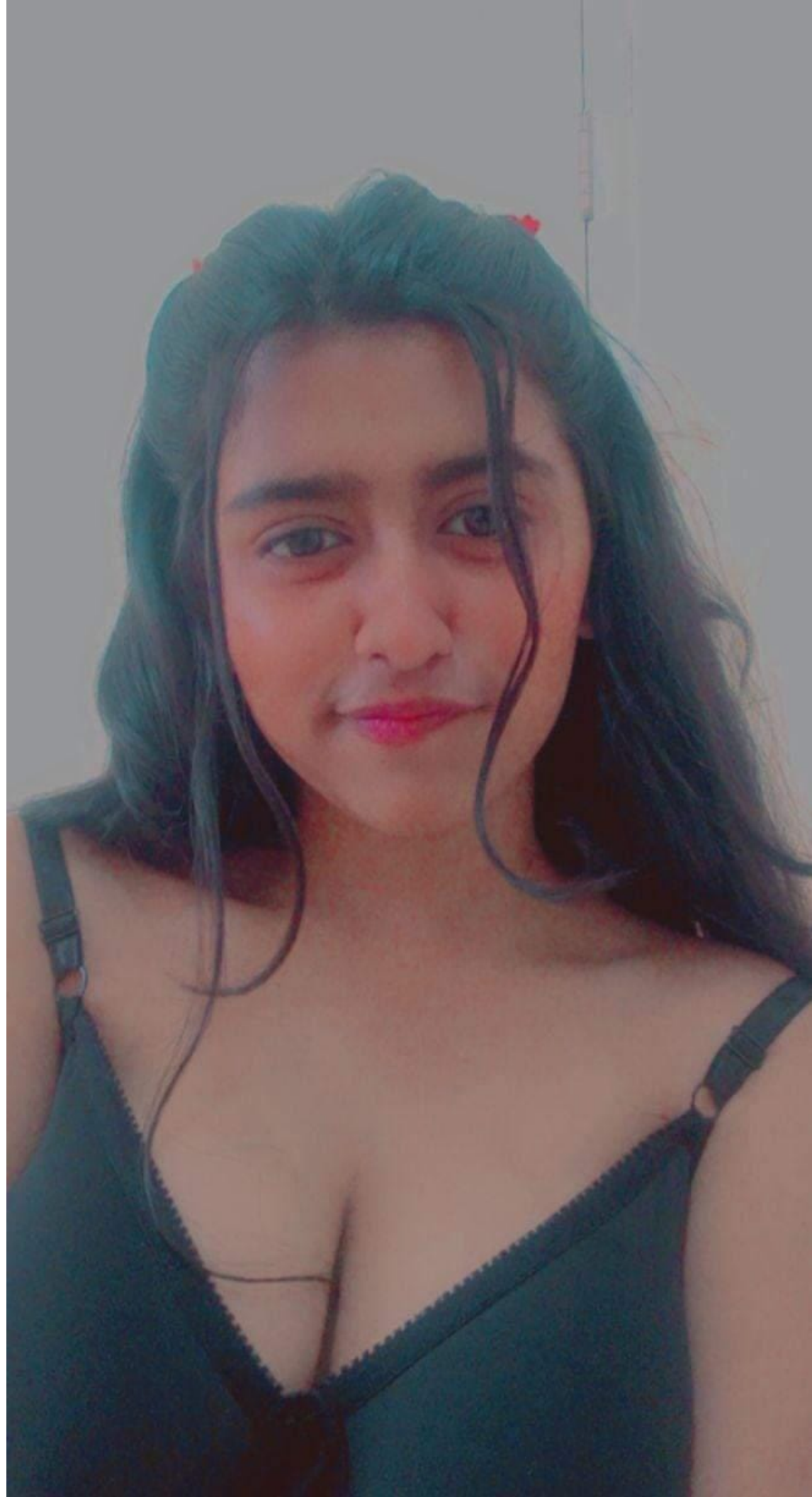


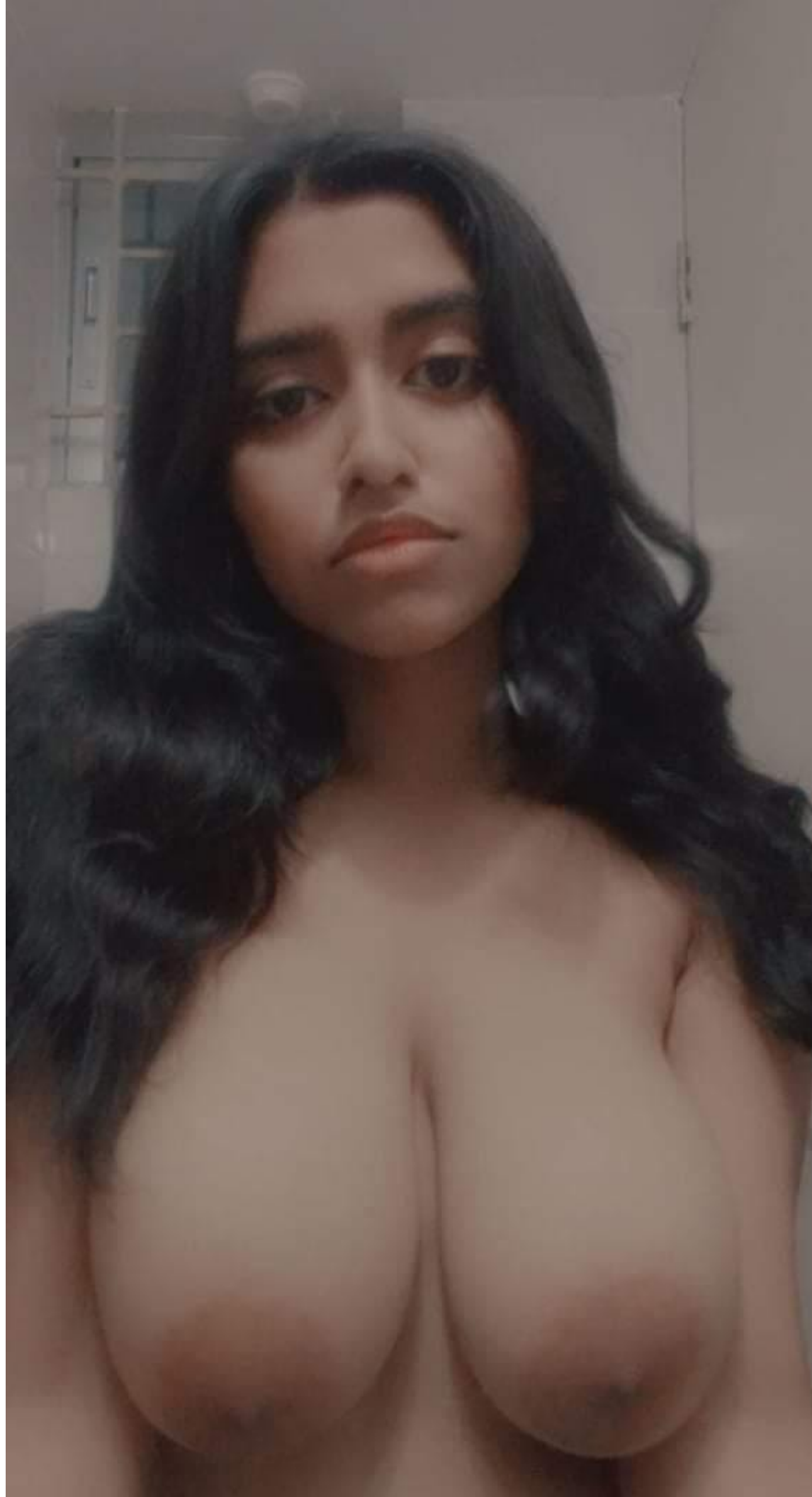






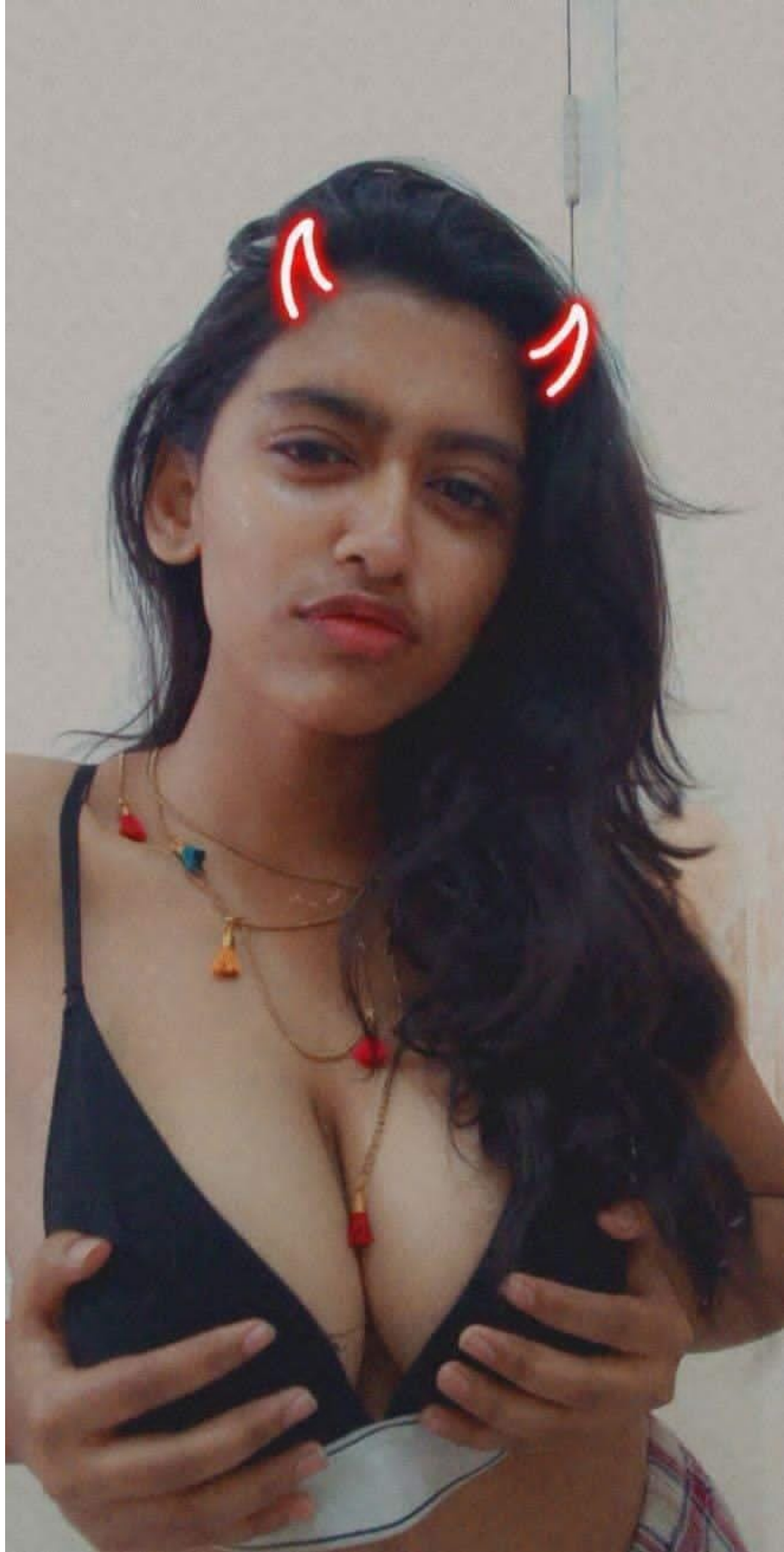




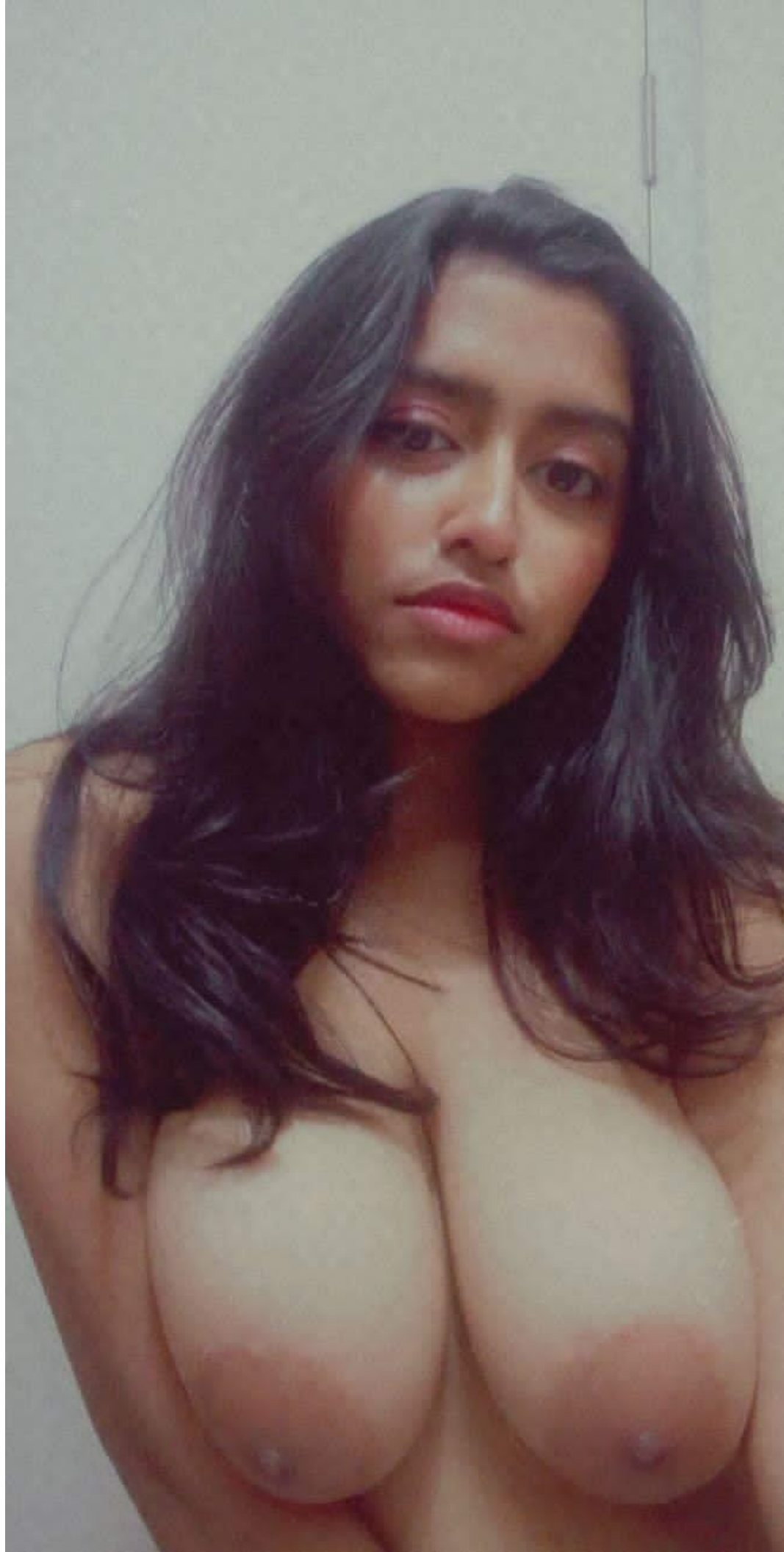




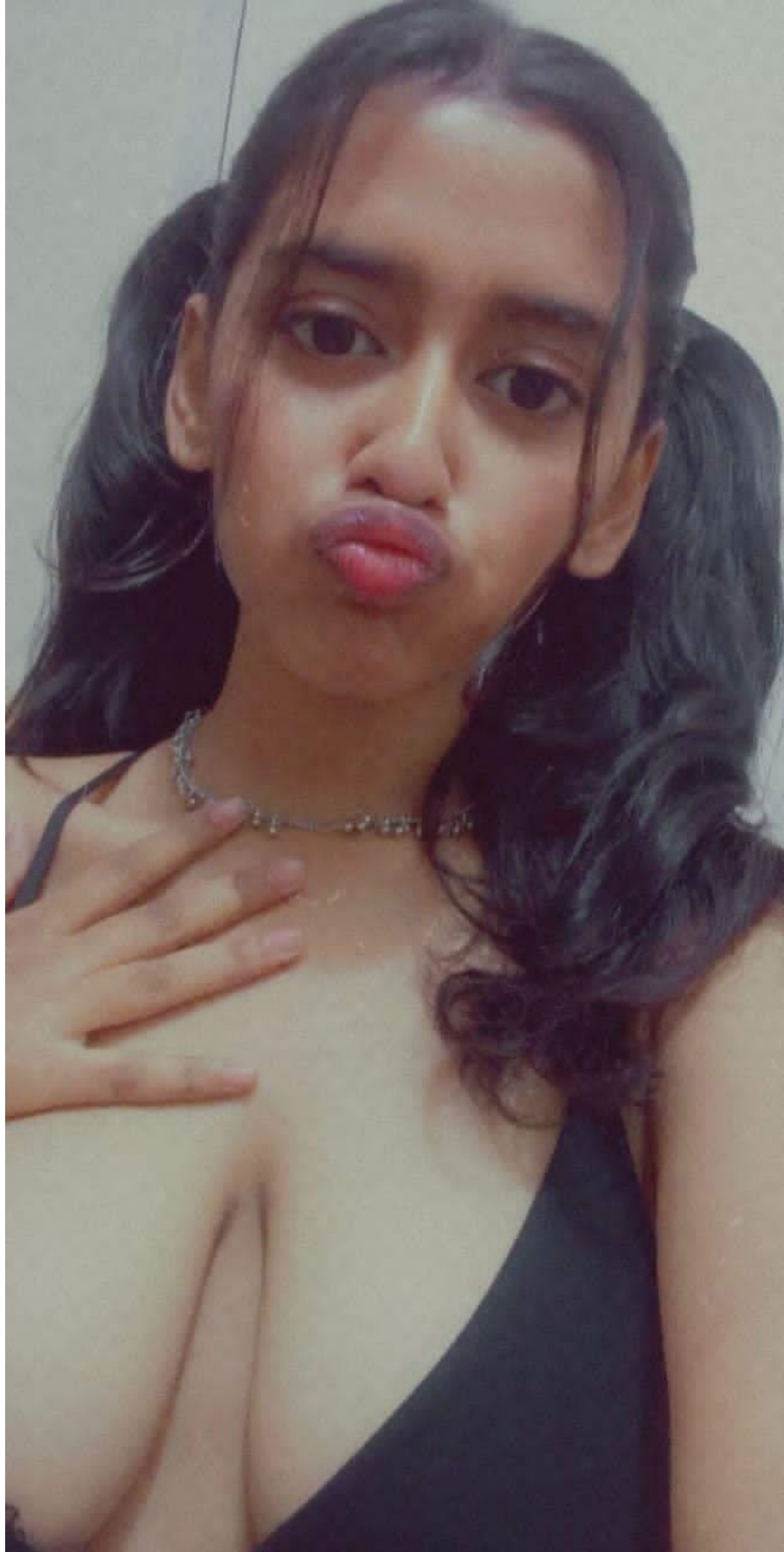


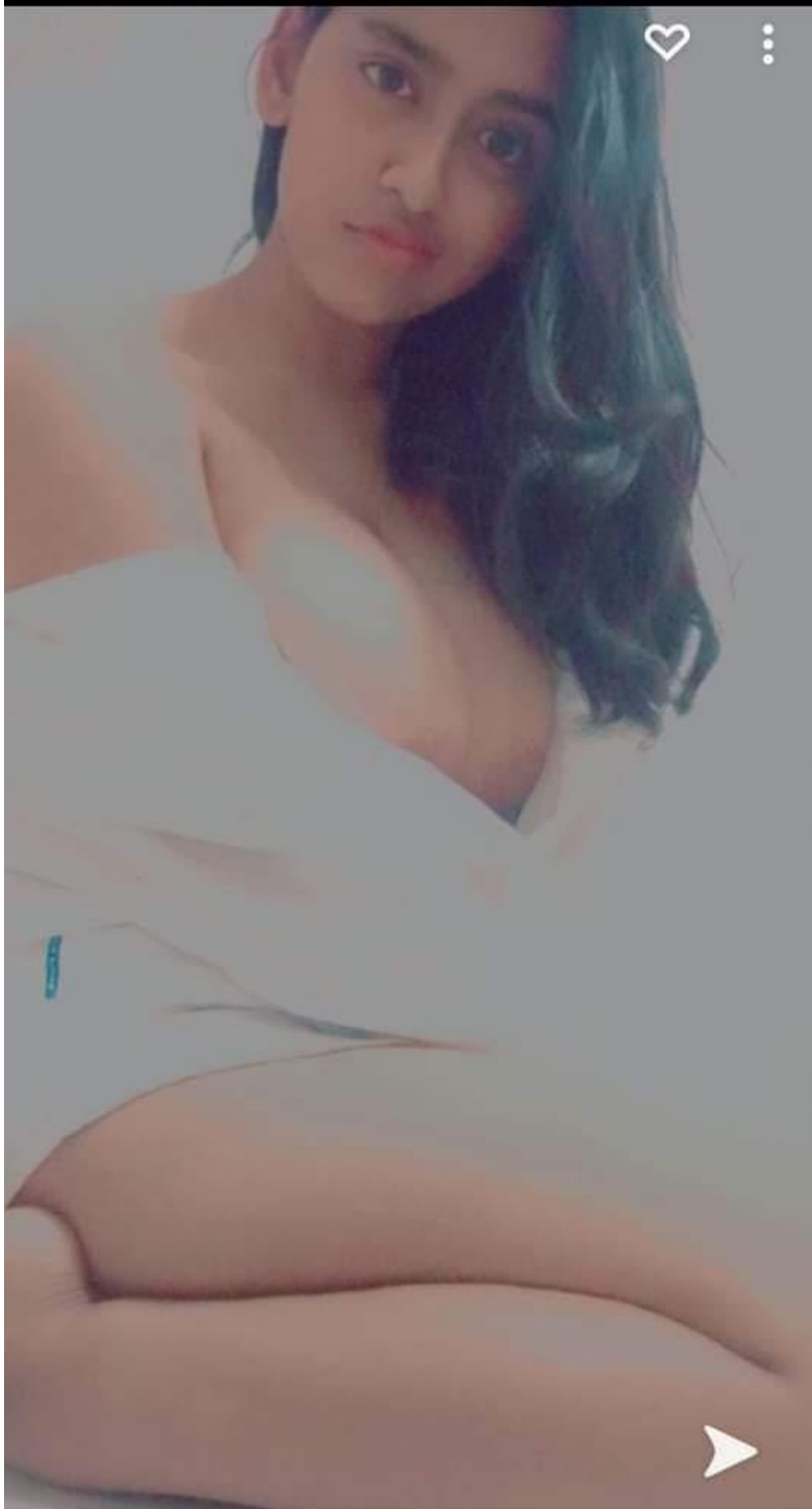




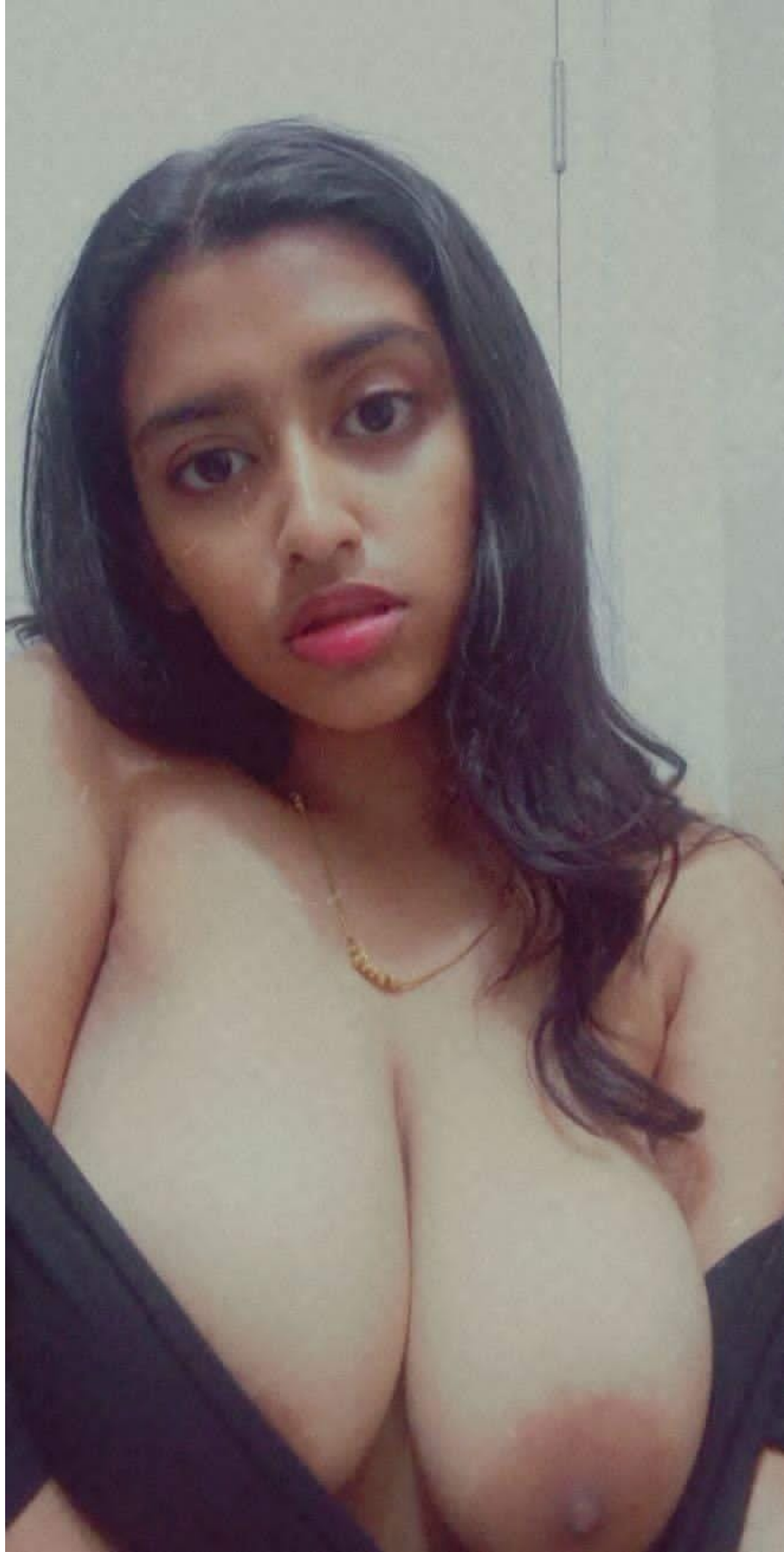


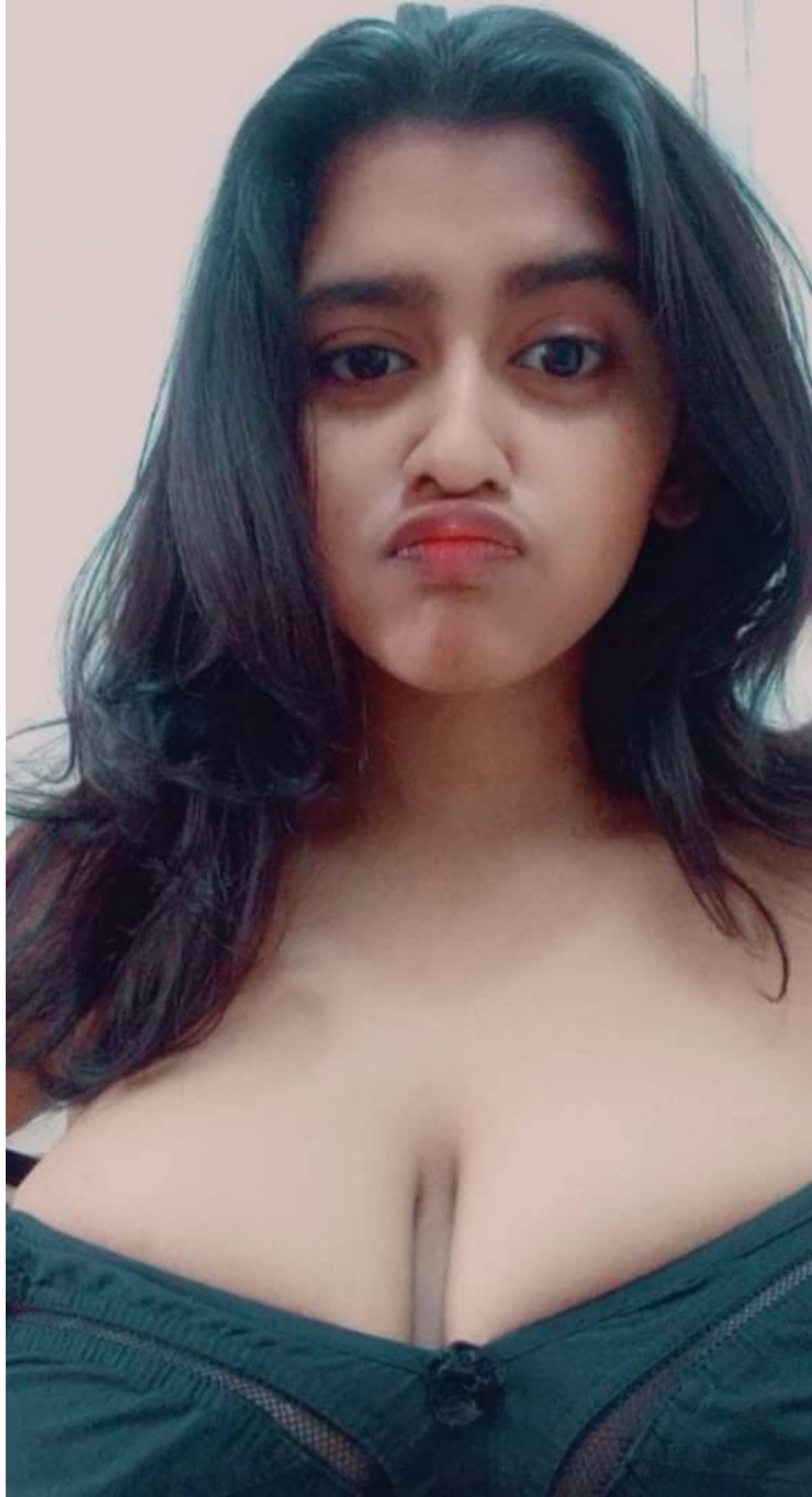




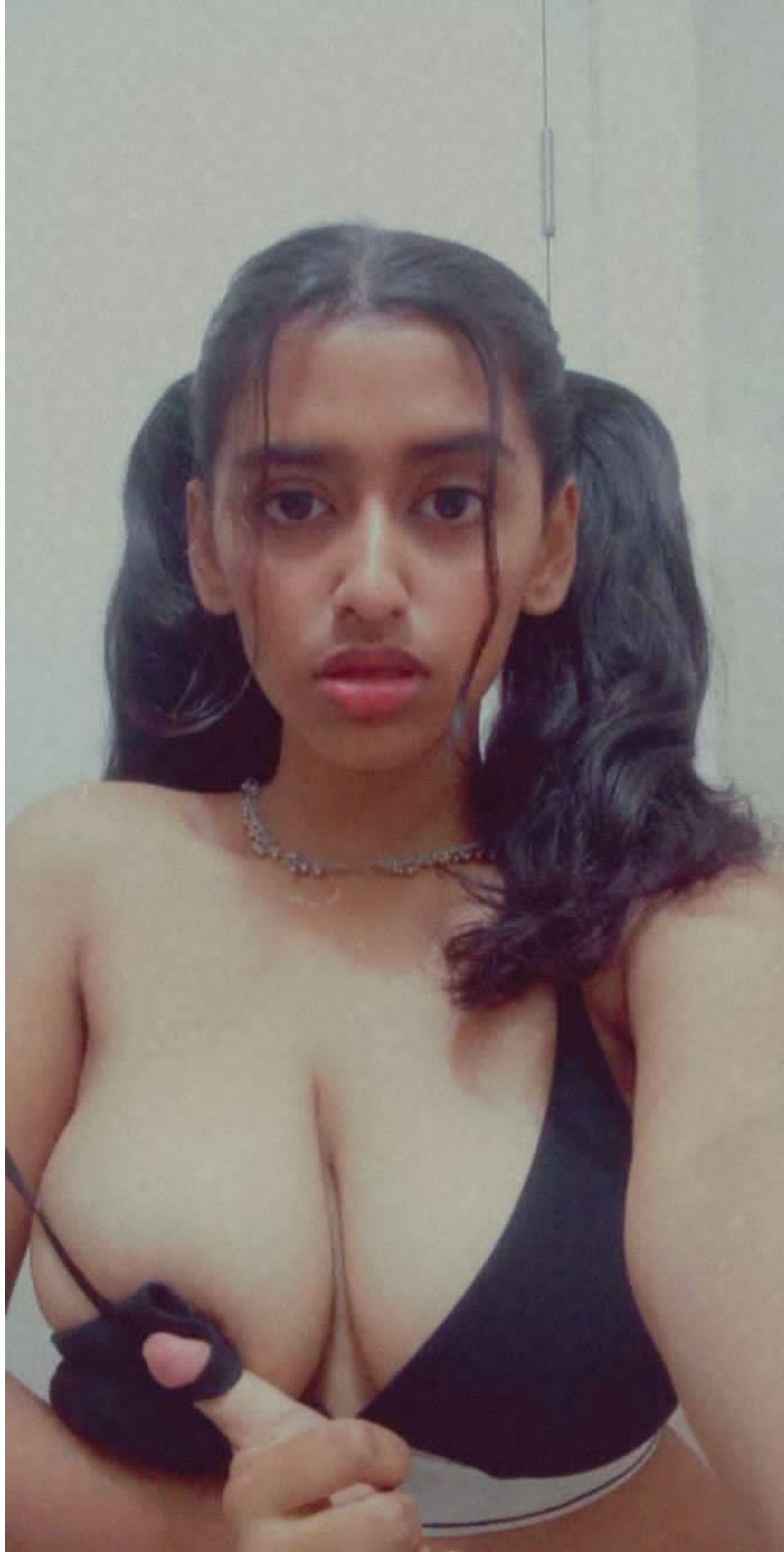






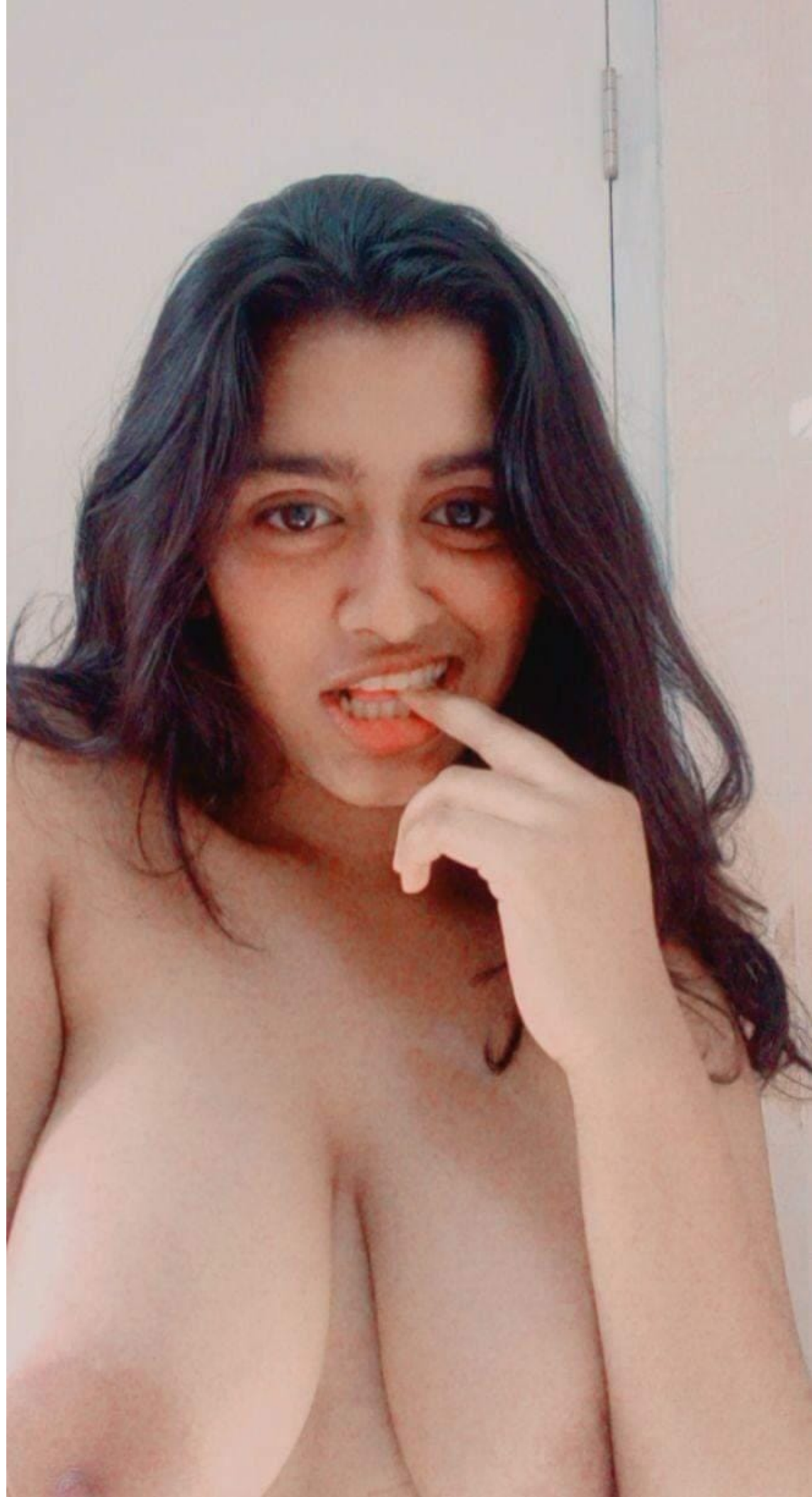




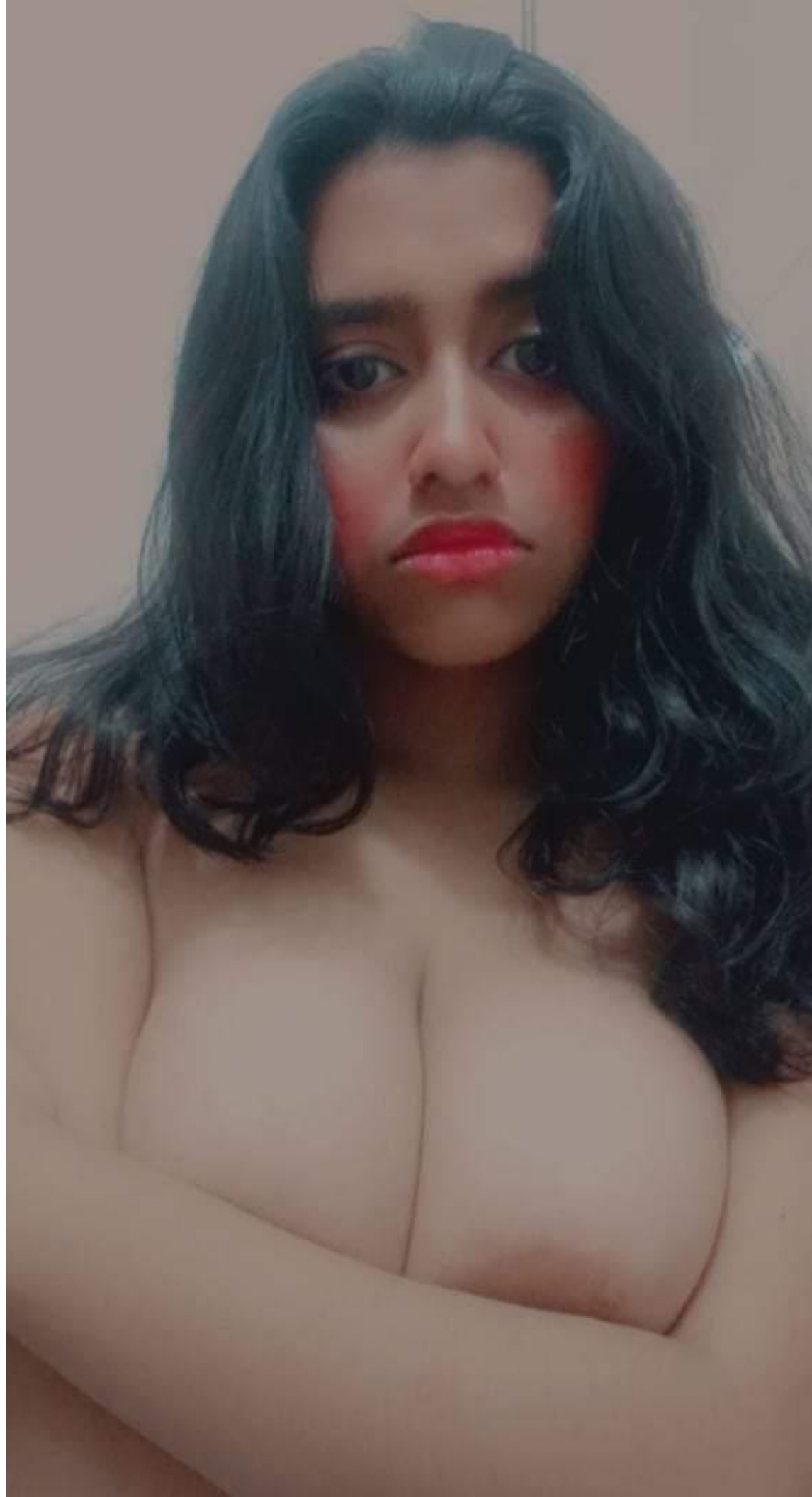


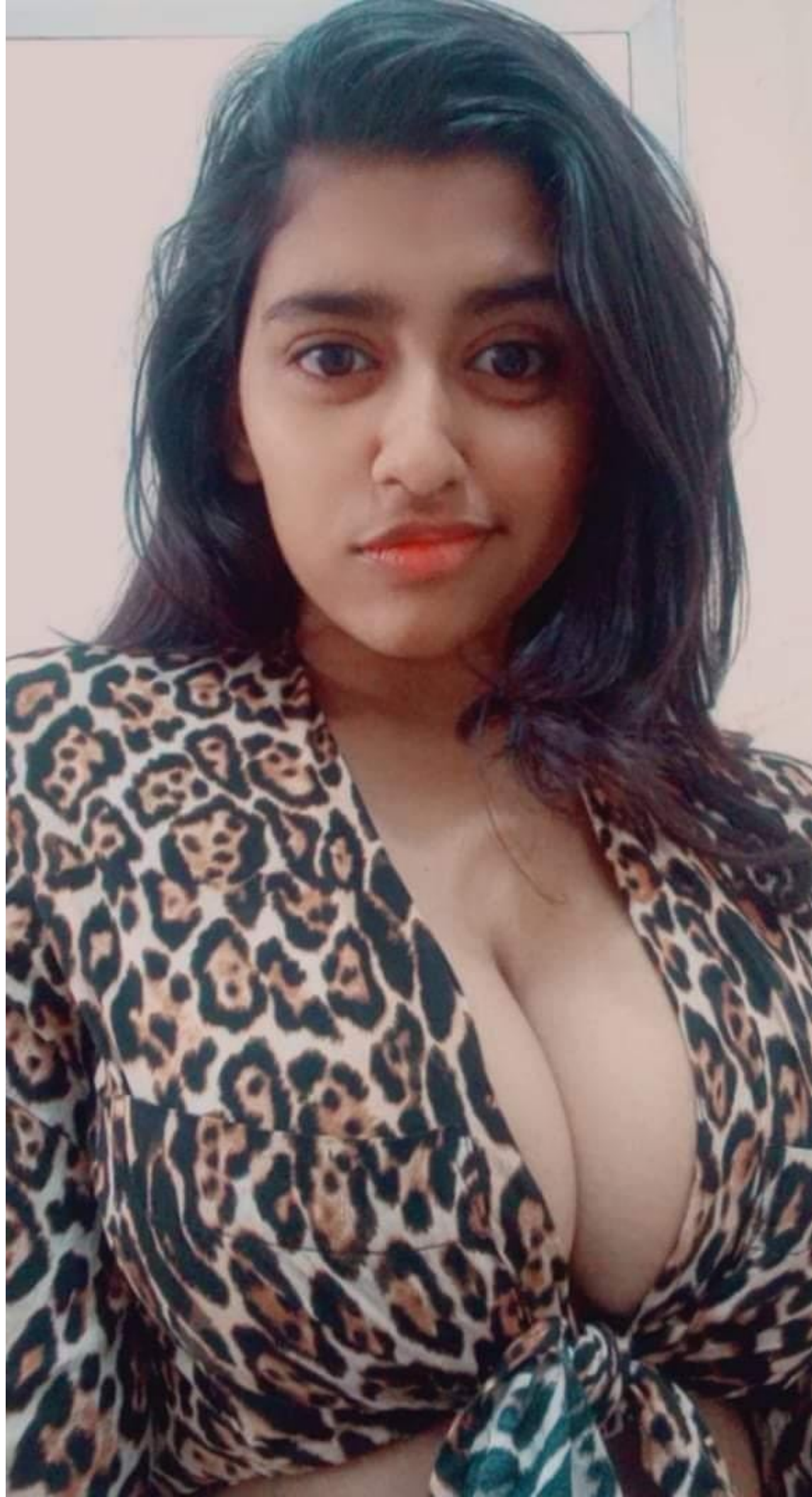


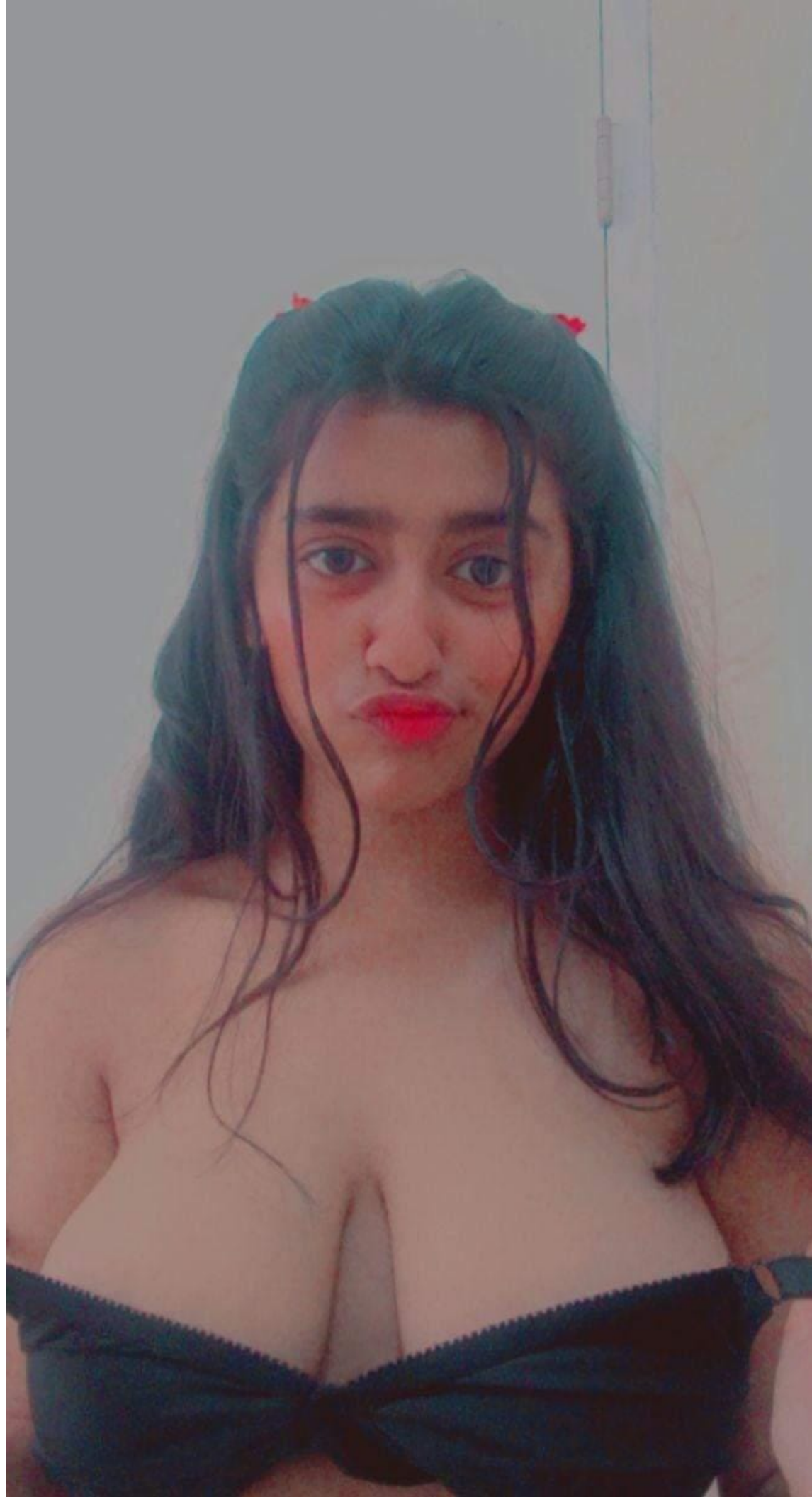




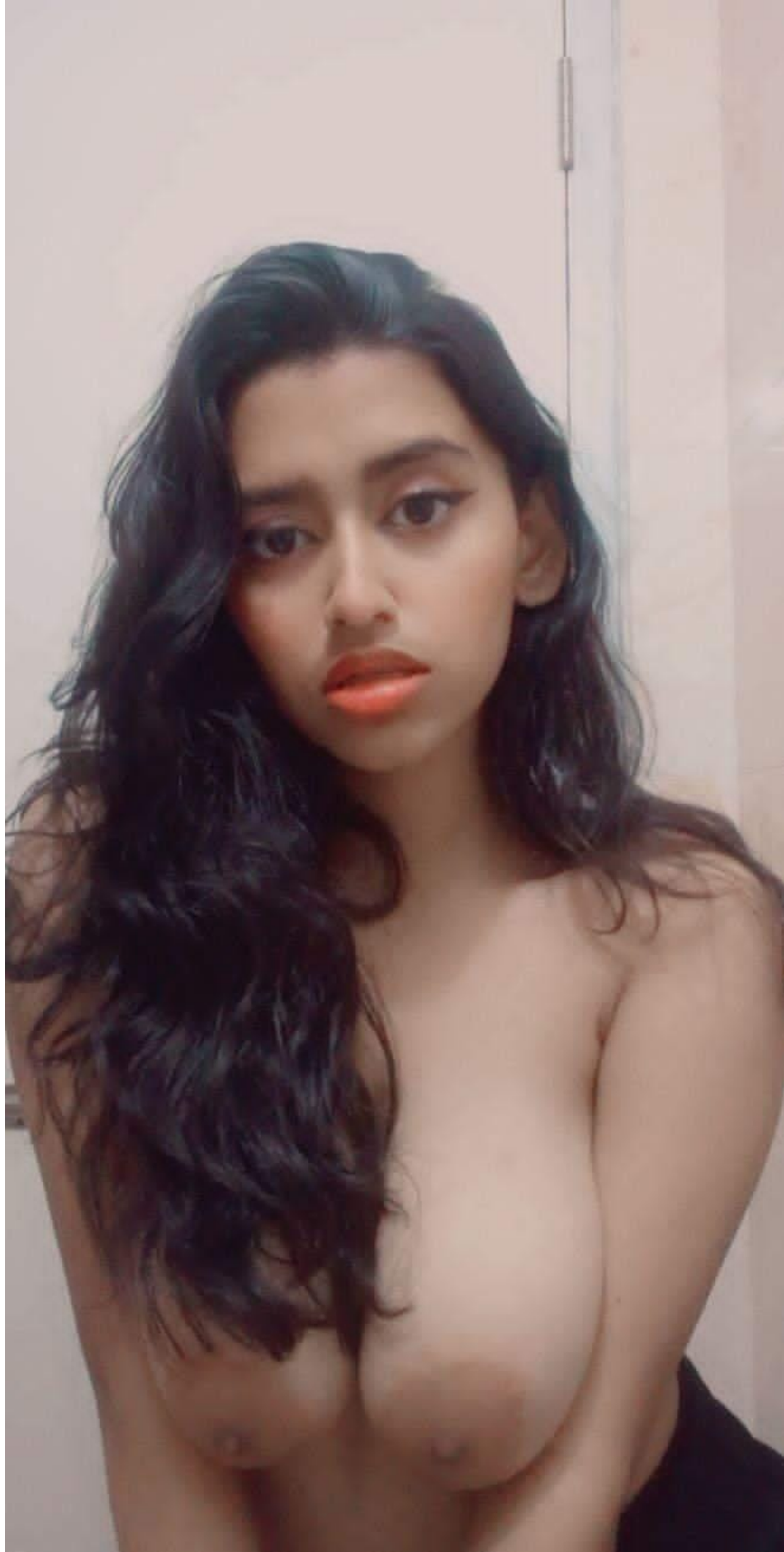




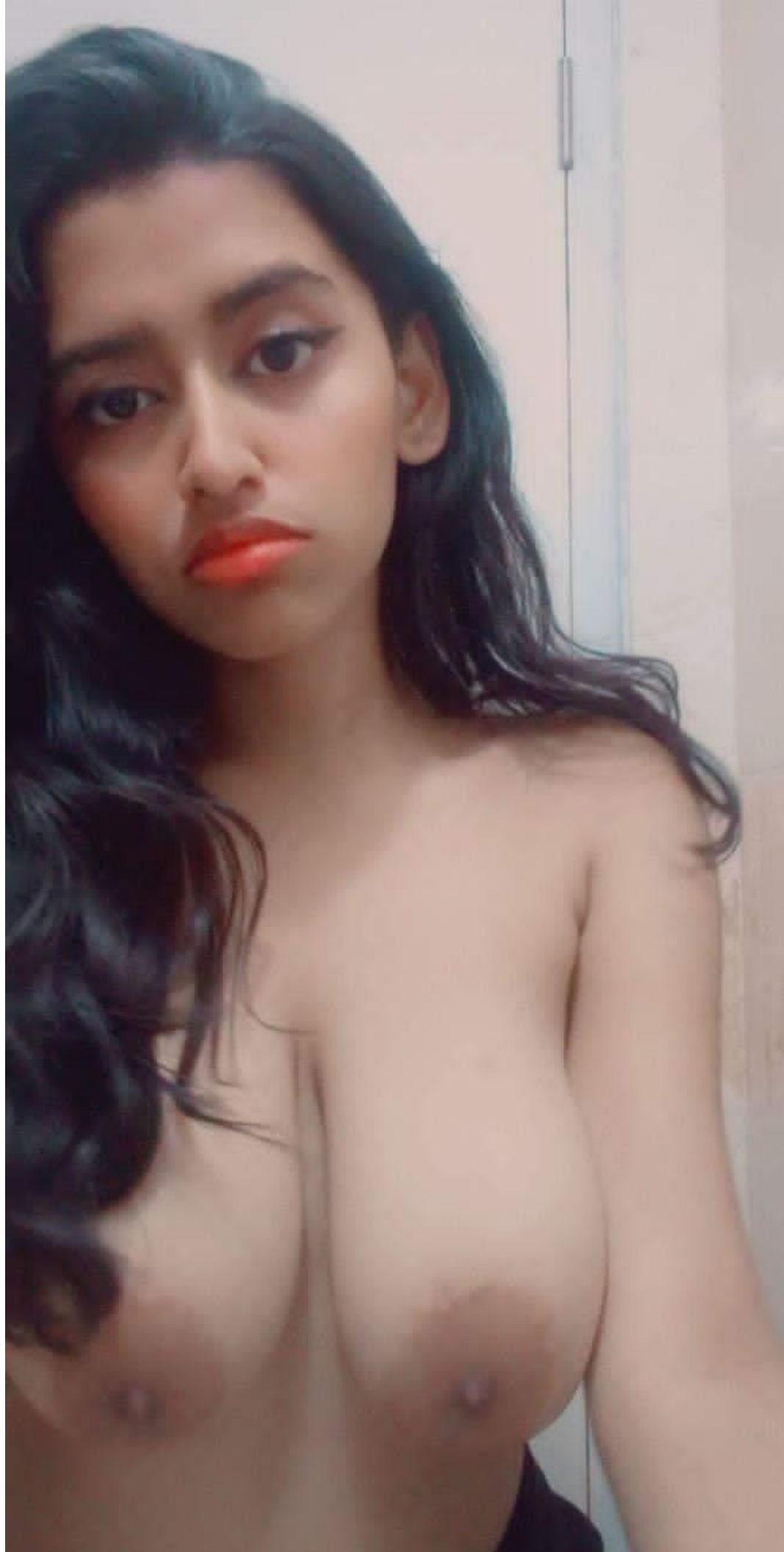


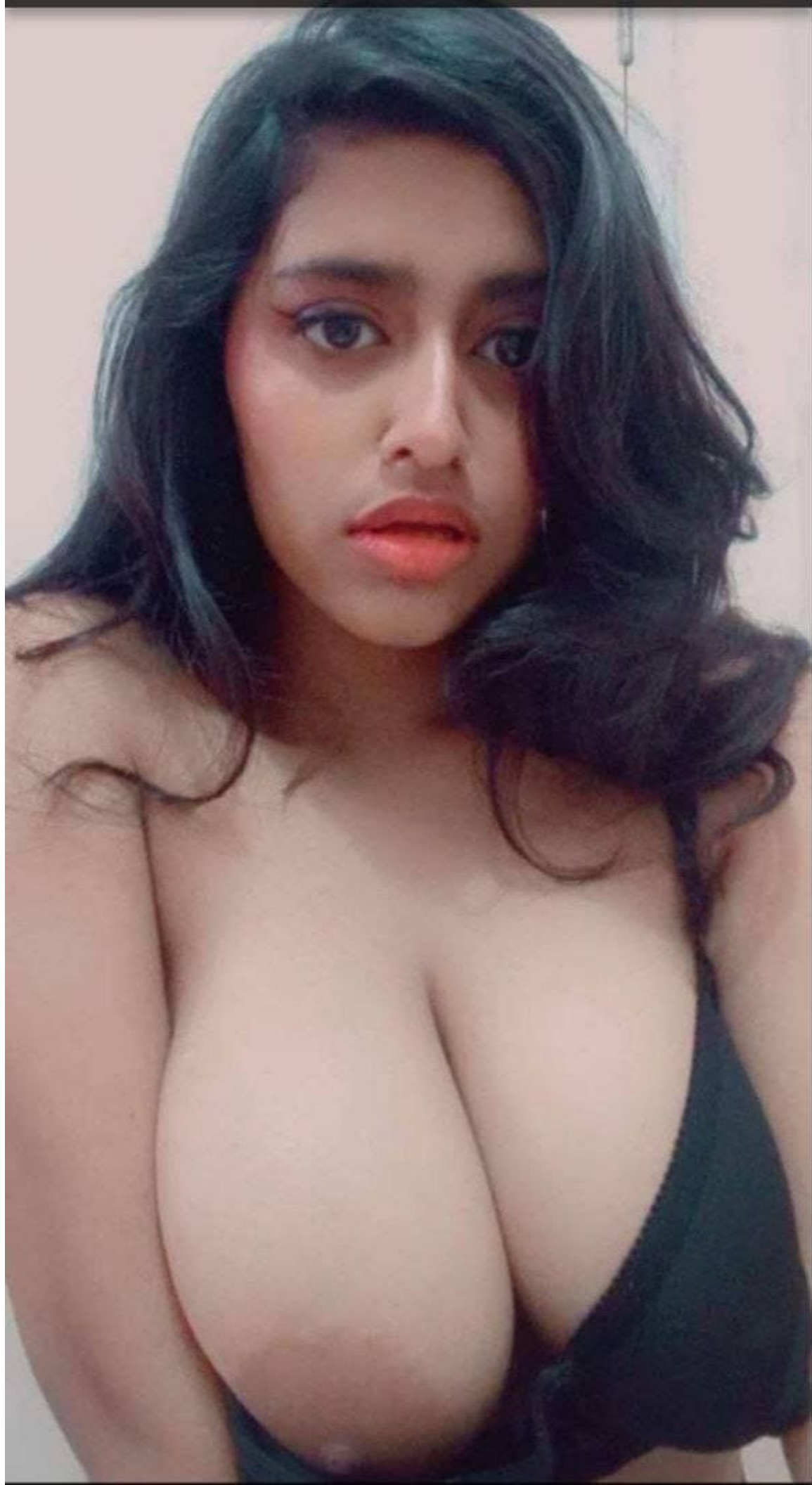


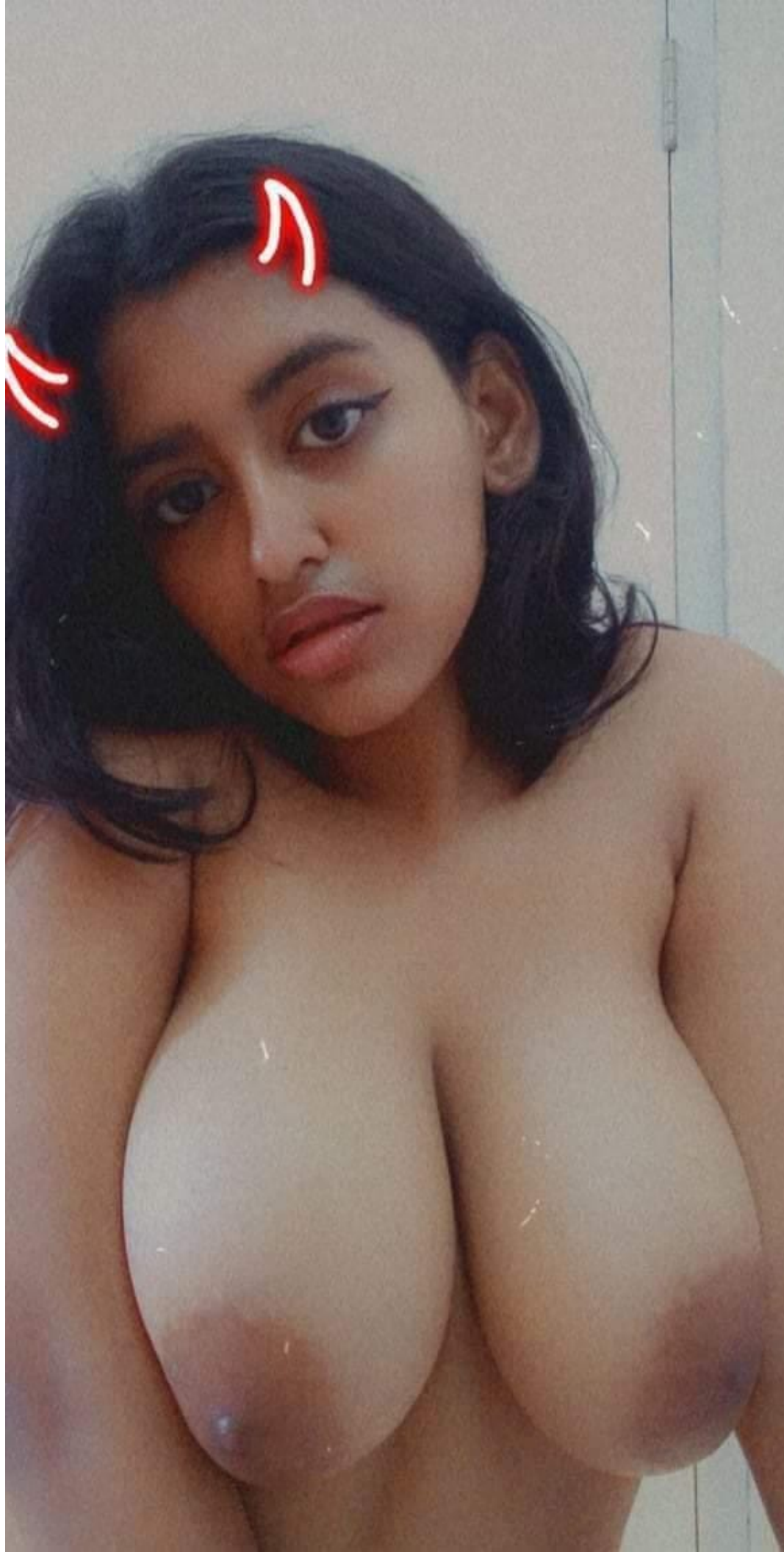






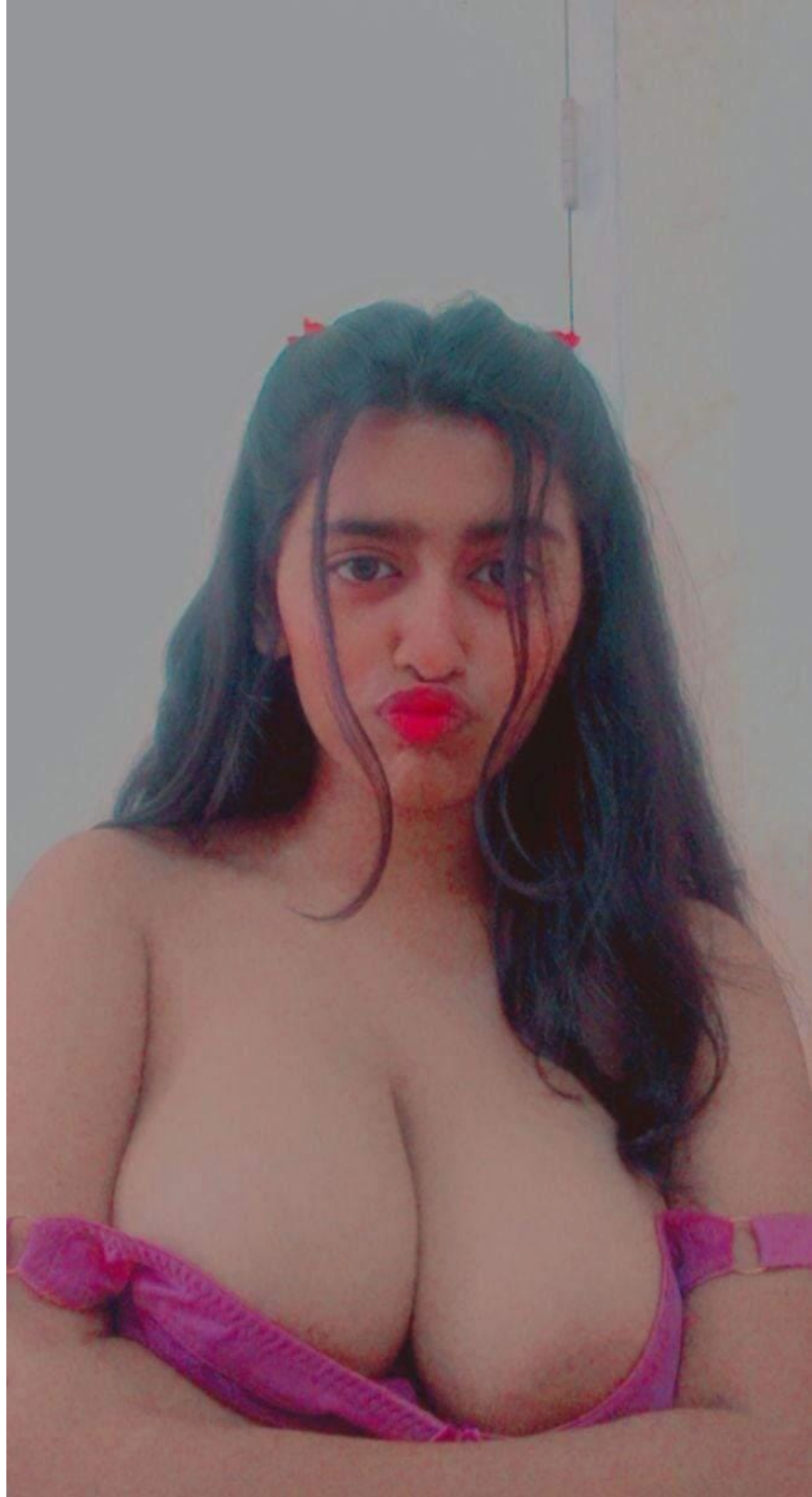


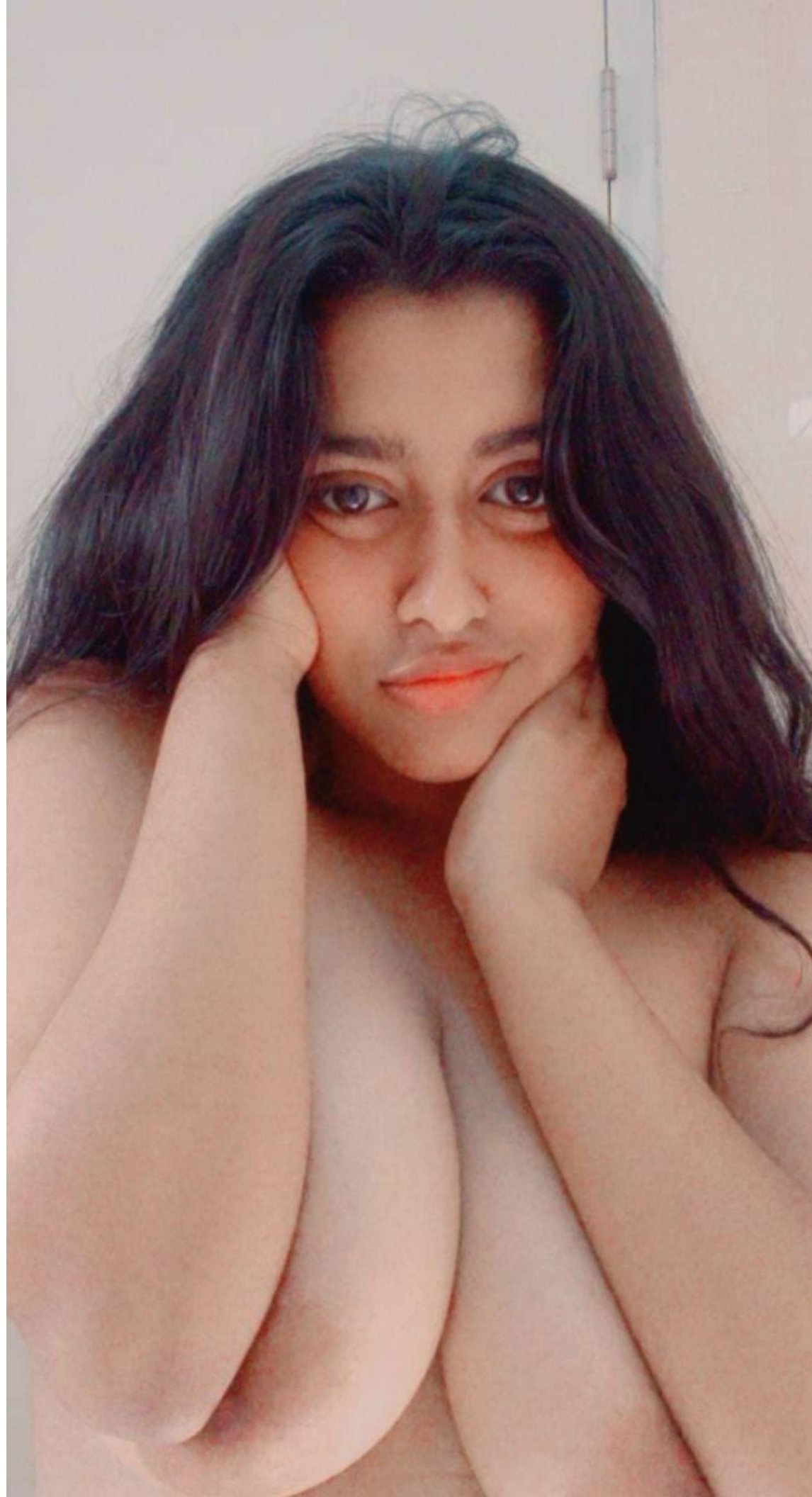


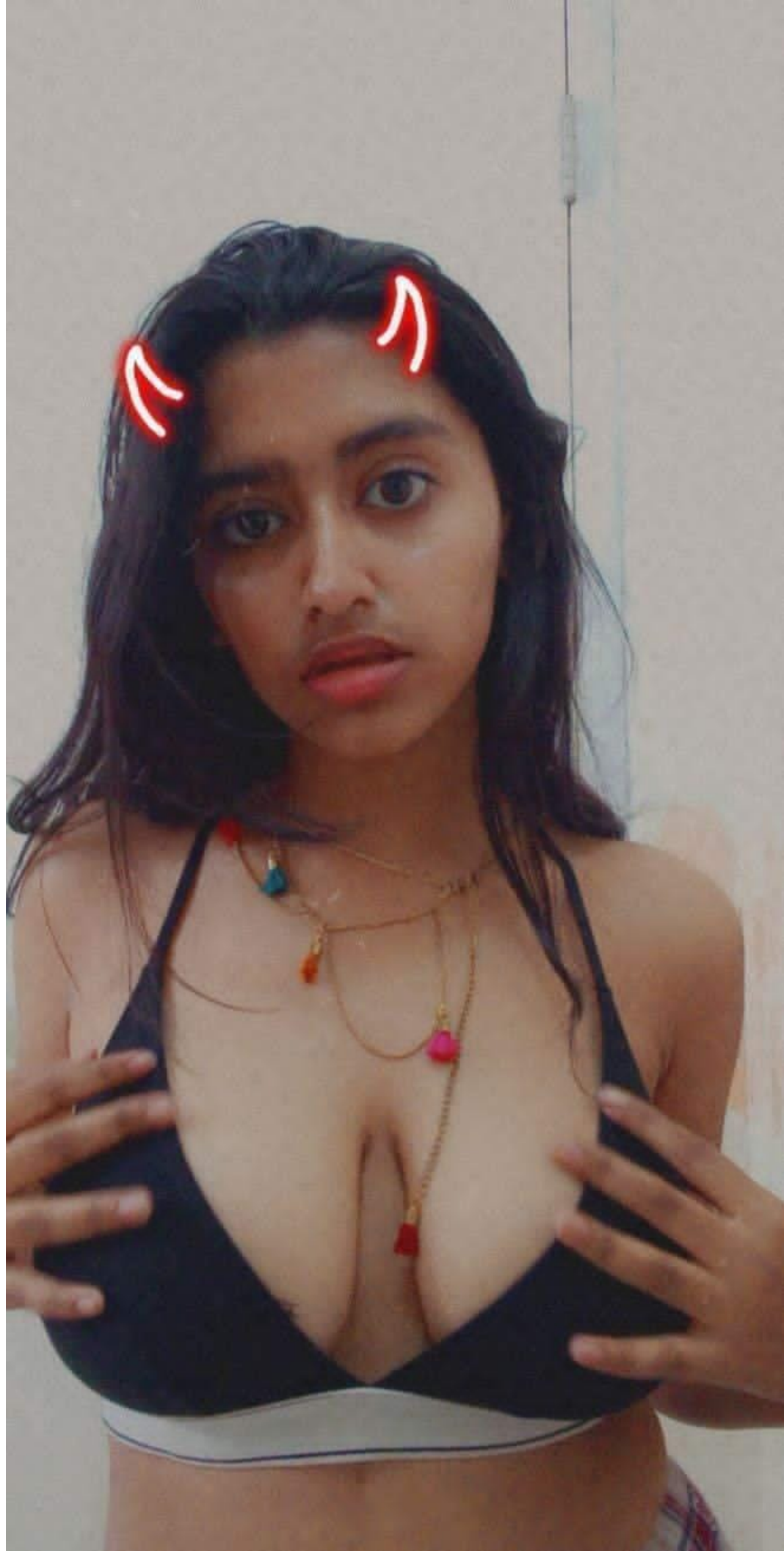
















C I A O







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